Direct Method

When direct method is used, the formula for finding the arithmetic mean is

$$\bar{X} = \frac{\sum fx}{\sum f}$$

Where x is the mid-point of the class interval and f is the corresponding frequency

Steps

(i) Obtain the mid-point of each class and denote it by x

(ii) Multiply those mid-points by the respective frequency of each class and obtain the sum of fx

(iii) Divide Σfx by Σf to obtain mean

Example 8.1 The following data gives the number of residents in an area based on their age. Find the average age of the residents.

Age	0-10	10-20	20-30	30-40	40-50	50-60
Number of Residents	2	6	9	7	4	2

Solution

Age	Number of Residents(f)	Midvalue(x)	fx
0-10	2	5	10
10-20	6	15	90
20-30	9	25	225
30-40	7	35	245
40-50	4	45	180
50-60	2	55	110
	$\sum f = 30$		$\sum fx = 860$

Mean= $\bar{X} = \frac{\sum fx}{\sum f} = \frac{860}{30} = 28.67$

Hence the average age $= 28.67$.

Assumed Mean Method

We have seen how to find the arithmetic mean of a grouped data quickly using the

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